From Policy Leadership to a Forward-Looking Monetary–Fiscal Game

Static intuition, recursive strategic investment, and a New Keynesian equilibrium

Monetary–Fiscal Coordination Project

2 October 2026

Build the model in three layers

1. Static game

What is policy conflict?

- simultaneous Nash;
- imposed Stackelberg order;
- cooperation;
- demand versus supply shocks.

2. Recursive game

Where can leadership come from?

- adjustment costs;
- inherited instruments;
- Markov feedback;
- strategic investment.

3. Forward-looking NK

How does the private sector change it?

- expected future output;
- expected inflation;
- policy/forecast fixed point;
- calibration last.

The final model should be understood as an **analytical extension**, not as a separate black box.

The question

Monetary and fiscal authorities affect the same economy, but can have different mandates and different instruments.

Mechanical interaction

Both policies affect output, inflation, and public debt.

Objective conflict

Monetary policy is relatively inflation-oriented; fiscal policy is relatively output-oriented.

Strategic question

Can an authority deliberately choose a persistent policy position so that its future self—and the rival authority—must respond differently?

Coordination question

When do individually optimal commitment attempts make joint outcomes worse?

Start with the smallest possible macroeconomic block

Collect the inflation and output gaps in $z = (\pi, x)'$. Let

$$z = \bar{z} + D_M m + D_F f + \varepsilon,$$

where m is the monetary instrument and f the fiscal instrument.

Transmission vectors

D_M and D_F contain the signed effects of each instrument on inflation and output. They need not point in the same direction.

Shock vector

ε changes the policy trade-off:

- a demand shock moves output and inflation together;
- a supply shock moves them in opposite directions.

Purpose of this block

It isolates the game. There are no expectations, dynamics, or debt yet.

Different mandates create different preferred policy mixes

For $i \in \{M, F\}$, let

$$L_i = (z - z_i^*)' W_i (z - z_i^*) + r_i a_i^2,$$

where $a_M = m$ and $a_F = f$.

Agency mandates

- W_M: relatively high inflation weight;
- W_F: relatively high output weight;
- r_i: cost or conservatism in moving the own instrument.

True social welfare

$$L_S = (z - z_S^*)' W_S (z - z_S^*) + r_M^S m^2 + r_F^S f^2.$$

Cooperation minimizes L_S; it does not average L_M and L_F.

Simultaneous Nash: intersect the best responses

The monetary first-order condition gives

$$m = \alpha_M + R_{MF} f,$$
$$R_{MF} = -\frac{D'_M W_M D_F}{r_M + D'_M W_M D_M}.$$

Similarly,

$$f = \alpha_F + R_{FM} m,$$
$$R_{FM} = -\frac{D'_F W_F D_M}{r_F + D'_F W_F D_F}.$$

$$\begin{bmatrix} m^N \\ f^N \end{bmatrix} = \frac{1}{1-\Gamma} \begin{bmatrix} 1 & R_{MF} \\ R_{FM} & 1 \end{bmatrix} \begin{bmatrix} \alpha_M \\ \alpha_F \end{bmatrix}, \qquad \Gamma = R_{MF} R_{FM}.$$

First analytical statistic

Γ is the product of the two reaction slopes. A large objective conflict does little strategically if either reaction slope is close to zero.

What does the response product mean?

Strategic complements: $\Gamma > 0$

The reaction slopes have the same sign. A stronger action by one authority induces a stronger response by the other.

- mutual amplification;
- possible offset race;
- denominator $1 - \Gamma$ becomes important.

Strategic substitutes: $\Gamma < 0$

The reaction slopes have opposite signs. One authority's action reduces the other's desired action.

- free riding can arise;
- policy burdens shift across agencies;
- interaction tends to damp rather than amplify.

The shock changes the intercepts and can change the economically relevant part of each reaction function.

A two-stage game: Stackelberg leadership

"Two-stage" here means an order of moves within one policy problem—not two calendar dates.

1. Monetary leader chooses m.
2. Fiscal follower observes m and chooses

$$f = BR_F(m) = \alpha_F + R_{FM} m.$$

3. The leader solves

$$\min_m L_M\big(m, BR_F(m)\big).$$

What changes relative to Nash?

The leader recognizes

$$\frac{dz}{dm} = D_M + D_F R_{FM}.$$

Nash treats f as fixed in its own first-order condition. The Stackelberg leader internalizes the follower's response.

Interchanging M and F gives the fiscal-leader game.

Demand and supply shocks produce different games

Demand disturbance

Output and inflation move together.

- Both authorities broadly want stabilization in the same direction.
- Each would prefer the other to bear more instrument cost.
- The central problem can be **free riding** and insufficient joint action.

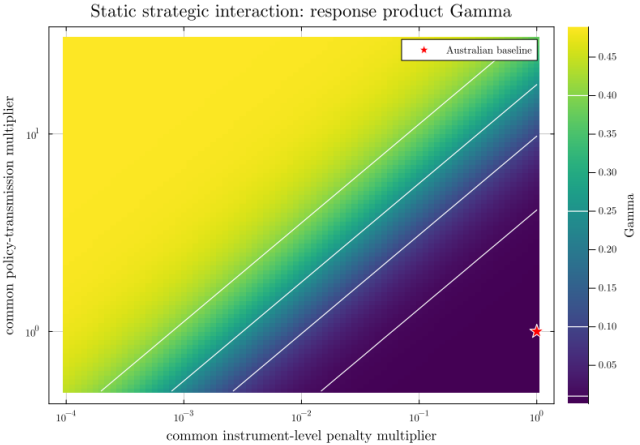
Cost-push disturbance

Inflation and output move in opposite directions.

- Different mandate weights imply different desired trade-offs.
- One authority may offset the macroeconomic consequences of the other.
- The central problem can be an **offset race** and excessive instrument movement.

Shock dependence is a property of the payoff cross-derivatives, not just a label attached to the innovation.

Static interaction is strong only in particular regions



- Stronger common transmission steepens the two best responses.
- Larger direct instrument penalties flatten them.
- The Australian baseline has a small static response product.

What the static game teaches—and what it cannot

It teaches

- why mandates alone are insufficient;
- how reaction slopes govern amplification;
- why the shock changes the strategic relationship;
- how imposed leadership differs from Nash.

It cannot explain

- where move order comes from;
- how today's policy constrains tomorrow;
- why leadership might emerge endogenously;
- how expected future policy affects current private behaviour.

We next replace **imposed timing** with an **intertemporal state**.

The two-date bridge

Now use two actual dates, $t = 1, 2$. Add an adjustment cost

$$\lambda_i (a_{i,2} - a_{i,1})^2.$$

Date 2

Given inherited (m_1, f_1), both authorities choose (m_2, f_2) using feedback rules

$$m_2 = k_M' s_2, \qquad f_2 = k_F' s_2.$$

Date 1

Each authority knows that $a_{i,1}$ changes the state inherited at date 2 and therefore changes both date-2 policy choices.

The adjustment cost converts a current policy choice into a **strategic investment in a future position**.

Open loop versus feedback in the two-date game

Open loop

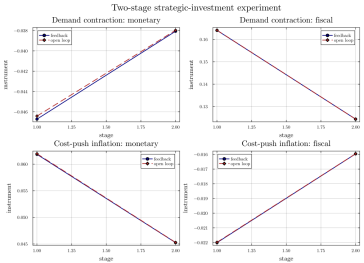
At date 1, future policy paths are treated as announced choices. A deviation in $a_{i,1}$ does not trigger a state-contingent revision in the rival's future action.

Feedback

At date 1, the authority internalizes

$$\frac{\partial a_{j,2}}{\partial a_{i,1}} = k_j' \frac{\partial s_2}{\partial a_{i,1}}.$$

This is the strategic-investment term that open loop omits.



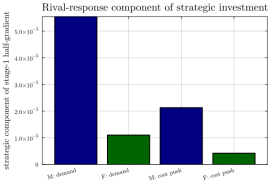
The strategic term is visible in the date-1 FOC

Write authority i's date-1 problem as

$$\min_{a_{i,1}} \ell_{i,1} + \beta_i V_{i,2}(s_2).$$

The continuation derivative contains

$$\frac{dV_{i,2}}{da_{i,1}} = \underbrace{V_{i,s}\frac{\partial s_2}{\partial a_{i,1}}}_{\text{own-state effect}} + \underbrace{V_{i,a_j}\frac{\partial a_{j,2}}{\partial s_2}\frac{\partial s_2}{\partial a_{i,1}}}_{\text{rival-response effect}}.$$



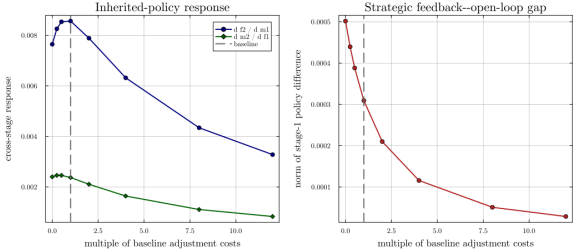
Interpretation

Adjustment cost is not itself the strategic effect. It activates a state; the rival's response to that state generates the strategic effect.

Why the adjustment-cost effect is generally hump-shaped

Ignoring cross-effects, own policy persistence behaves approximately like

$$k_{i,a_{i,-1}} \approx \frac{\lambda_i}{r_i + \lambda_i + \beta_i B_i' P_i B_i}.$$



- $\lambda_i = 0$: no inherited-policy channel.
- Moderate λ_i: current policy binds the future and can move the rival.
- Very large λ_i: policy barely moves today, so investment capacity falls.

Lesson

More inertia is not monotonically more leadership.

The infinite-horizon recursive game

Let the predetermined state contain macro states, inherited instruments, and shocks:

$$s_t = (q_t, m_{t-1}, f_{t-1}, \varepsilon_t)'.$$

The agency Bellman equations are

$$V_M(s_t) = \min_{m_t} \left\{ \ell_M(s_t, m_t, f_t) + \beta_M V_M(s_{t+1}) \right\},$$
$$V_F(s_t) = \min_{f_t} \left\{ \ell_F(s_t, m_t, f_t) + \beta_F V_F(s_{t+1}) \right\}.$$

Conjecture linear Markov rules and quadratic values:

$$m_t = k_M' s_t, \, qquad f_t = k_F' s_t, \, qquad V_i(s_t) = s_t' P_i s_t.$$

Endogenous leadership

No authority moves first within the period. Leader-like behaviour can emerge only because current policy changes the state to which future players respond.

Four non-cooperative commitment environments

Game	State created by adjustment cost	Interpretation
Flexible Nash	none	no inherited-policy commitment
Monetary persistence	m_{t-1}	unilateral monetary capacity
Fiscal persistence	f_{t-1}	unilateral fiscal capacity
Bilateral persistence	m_{t-1}, f_{t-1}	both invest in positions

Coordination benchmark

A discretionary planner chooses both instruments under a distinct social loss. Coordination changes the objective; persistence changes the policy technology.

Not yet a literal meta-game

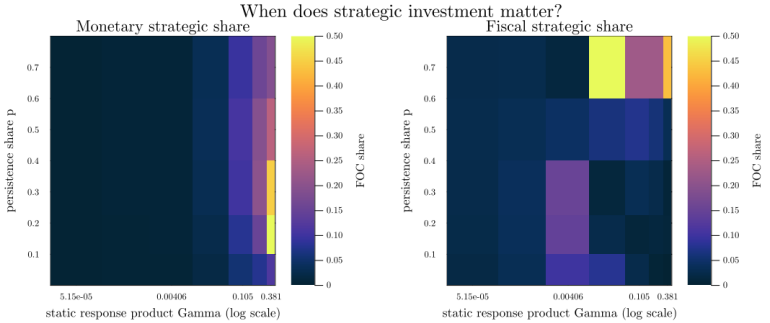
λ_M and λ_F are imposed primitives. A true outer Nash game would let institutions choose those commitment technologies first.

The initial recursive model: where should strategy matter?

Two dimensionless statistics summarize the mechanism:

$$\Gamma = R_{MF} R_{FM} \qquad \text{static cross-response product,}$$

$$p_i = \frac{\lambda_i}{D_i + \lambda_i} \qquad \text{adjustment share of own curvature.}$$



- Persistence supplies the state; Γ supplies the rival response.
- Both are required for a large strategic share.

What is still missing from the initial recursive model?

What it gets right

- feedback rather than open-loop strategies;
- inherited instruments as endogenous states;
- strategic investment in rival responses;
- infinite-horizon policy choice.

What it abstracts from

- households looking forward when choosing demand;
- firms looking forward when setting prices;
- current outcomes responding to the expected future policy path;
- a rational-expectations fixed point.

The next step is to keep the policy game but replace the private block.

Step 1: replace ad hoc demand dynamics with an NK IS curve

A log-linear household Euler equation motivates

$$x_t = \mathbb{E}_t x_{t+1} - \sigma\left(m_t - \mathbb{E}_t \pi_{t+1}\right) + \psi f_t + d_t.$$

Direct effects

- monetary tightening lowers current demand: $-\sigma m_t$;
- fiscal expansion raises it: $+\psi f_t$;
- d_t is a demand disturbance.

Forward-looking effect

Expected future output and the expected real policy stance affect demand today. An anticipated future policy response can move x_t before that policy is implemented.

Step 2: add forward-looking price setting

Calvo-style price setting gives the New Keynesian Phillips curve

$$\pi_t = \beta_p \mathbb{E}_t \pi_{t+1} + \kappa x_t + u_t.$$

Current marginal cost

The output gap raises inflation through κx_t.

Expected future inflation

Firms that cannot reset prices every period care about the future price-setting environment. Expected future policy therefore enters current inflation through $\mathbb{E}_t \pi_{t+1}$.

The cost-push shock u_t creates the inflation–output trade-off that makes mandate asymmetry consequential.

Step 3: retain debt and the policy-investment states

Use the debt-gap law

$$b_{t+1} = \rho_b b_t + \chi_m m_t + \chi_f f_t - \chi_\pi \pi_t.$$

The minimal predetermined state is

$$s_t = (b_t, m_{t-1}, f_{t-1}, d_t, u_t)'.$$

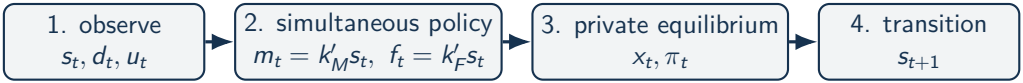
Macro strategic state

Debt persists even without adjustment costs. Both authorities can affect it and respond to it.

Petkov strategic state

m_{t-1} and f_{t-1} matter only when the corresponding adjustment cost is positive. They isolate the inherited-policy channel.

Step 4: specify timing carefully



- Private agents observe current policy and know the stationary rules used from $t+1$ onward.
- Policymakers understand how current instruments affect private outcomes and the next state.
- No Taylor rule is imposed: both instruments are solutions to agency optimization problems.

Step 5: conjecture a private forecasting rule

Let $y_t = (x_t, \pi_t)'$ and $q_t = (s_t', m_t, f_t)'$. Before imposing equilibrium, write

$$s_{t+1} = Nq_t + Ey_t.$$

Private agents conjecture

$$\boxed{\mathbb{E}_t y_{t+1} = Gs_{t+1}.}$$

What G means

Each column says how a predetermined state—debt, inherited instruments, or a shock—changes expected future output and inflation under the anticipated policy rules.

Why this is the new fixed point

The policy rules help determine G, but policy optimization itself depends on the private responses generated by G.

Step 6: reduce the private block conditional on beliefs

Collect the IS and Phillips curves as

$$Ly_t = Dq_t + HGs_{t+1}, qquadL = \begin{bmatrix} 1 & 0 \\ -\kappa & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & \sigma \\ 0 & \beta_p \end{bmatrix}.$$

Substitute $s_{t+1} = Nq_t + Ey_t$:

$$\begin{aligned} [L - HGE]y_t &= [D + HGN]q_t, \\ y_t &= O(G)q_t, \\ O(G) &= [L - HGE]^{-1}[D + HGN]. \end{aligned}$$

The private expectations multiplier

$[L - HGE]^{-1}H$ measures how strongly expected future outcomes feed back into current output and inflation.

Step 7: the policy game is LQ conditional on G

Partition the reduced transition:

$$s_{t+1} = A(G)s_t + B_M(G)m_t + B_F(G)f_t.$$

The stage loss becomes $q_t' Q_i(G) q_t$, and $V_i(s) = s' P_i s$. Simultaneous FOCs can be written

$$\begin{bmatrix} Q^M_{mm} + \beta_M B_M' P_M B_M & Q^M_{mf} + \beta_M B_M' P_M B_F \\ Q^F_{fm} + \beta_F B_F' P_F B_M & Q^F_{ff} + \beta_F B_F' P_F B_F \end{bmatrix} \begin{bmatrix} m_t \\ f_t \end{bmatrix} = - \begin{bmatrix} Q^M_{ms} + \beta_M B_M' P_M A \\ Q^F_{fs} + \beta_F B_F' P_F A \end{bmatrix} s_t.$$

Key point

Conditional on private beliefs, this is the same recursive policy logic as the initial dynamic model. Forward-lookingness changes the reduced matrices and loss curvature through $O(G)$.

Step 8: close the rational-expectations fixed point

For a provisional G:

1. solve the coupled policy Bellman/Riccati equations;
2. obtain $m_t = k_M(G)'s_t$ and $f_t = k_F(G)'s_t$;
3. calculate the implied private outcome rule.

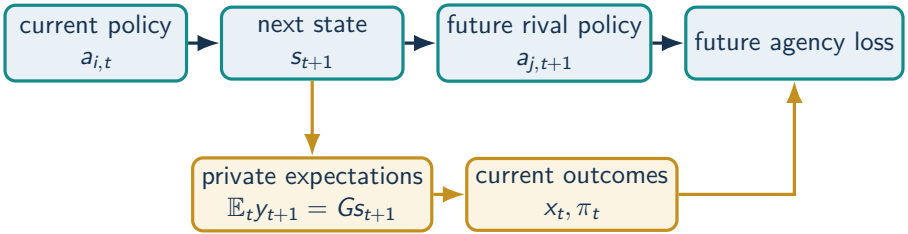
Partition $O(G) = (O_s, O_m, O_f)$. Rational expectations require

$$\boxed{G = \Phi(G) = O_s(G) + O_m(G)k_M(G)' + O_f(G)k_F(G)'.}$$

Equilibrium

A solution is simultaneously optimal for both policy agencies and validates the private sector's forecast of future output and inflation.

How forward-looking behaviour changes strategic investment



New channel, same necessary condition

Expectations transmit future policy into current outcomes. They amplify the rival response $k_j' B_i$; they cannot substitute for a weak rival response.

The strategic FOC now has three interpretable channels

For authority i,

$$\beta_i B_i' P_i s_{t+1} = \underbrace{\mathcal{N}_i}_{\text{own continuation}} + \underbrace{\mathcal{S}_i^{macro}}_{\text{rival response through debt/macro states}} + \underbrace{\mathcal{S}_i^{policy}}_{\text{rival response through inherited policy}} .$$

Macro state

Current policy changes debt; future rival policy responds to debt.

Policy state

Current policy becomes m_{t-1} or f_{t-1}; the rival responds to that inherited position.

Expectations

Both future responses alter current x_t and π_t through the NK equations.

The specifically Petkov-style object is $\mathcal{S}_i^{policy}$.

Analytical conditions for large effects

The strategic term is approximately a response product:

$$\beta_i \times \underbrace{B_i}_{\text{state exposure}} \times \underbrace{k_j}_{\text{rival slope}} \times \underbrace{\mathcal{H}_j^i}_{\text{payoff shadow value}} \times \underbrace{[L - HGE]^{-1}H}_{\text{expectations transmission}} .$$

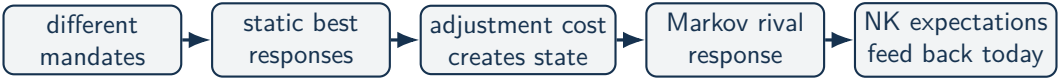
Larger effects are more likely with:

- moderate-to-high adjustment persistence;
- strong monetary/fiscal transmission;
- persistent debt or shocks;
- patient agencies.

But only if:

- the rival actually reacts to the inherited state;
- the reaction matters for the first authority's loss;
- the closed loop remains stable;
- current policy can still move.

The complete conceptual ladder



Necessary for conflict

Different preferred macroeconomic trade-offs and shared policy transmission.

Necessary for strategic investment

A current action changes a persistent state and the rival responds to it.

What the NK block adds

The future response affects current private decisions, potentially amplifying or attenuating the investment motive.

What it does not add automatically

Leadership, multiplicity, or strategic complementarity.

Calibration comes after the mechanism

The quantitative exercise asks three distinct questions:

1. Are policy and private-sector transmission coefficients large enough to generate steep cross-responses?
2. Do the calibrated adjustment costs create economically meaningful inherited-policy states?
3. Does the rational-expectations fixed point amplify those responses or stabilize them away?

Discipline

We report strategic FOC shares together with policy displacement and stability. A large share caused by cancellation in a tiny denominator is not treated as a large economic effect.

Australian quarterly calibration

Structural parameter	Value
σ	0.03695
ψ	0.05142
κ	0.15511
β_p	0.99
ρ_b	0.995
$(\chi_m, \chi_f, \chi_\pi)$	$(.015, .060, .100)$
(λ_M, λ_F)	$(.05, .04)$

Weights	Inflation	Output	Debt
Monetary	1.50	0.25	0.15
Fiscal	0.50	0.75	0.15
Social	1.00	0.50	0.15

Interpretation

Transmission coefficients are descriptive Australian estimates. Mandate weights and adjustment costs are transparent calibrations, not structurally identified parameters.

Data inputs: Reserve Bank of Australia statistical tables.

Impact policy responses

Demand contraction

Regime	m_0	f_0	Actual loss
Flexible Nash	-0.0874	-0.0843	0.1154
M persistence	-0.0660	-0.0802	0.1153
F persistence	-0.0883	-0.0648	0.1154
Bilateral	-0.0667	-0.0615	0.1153
Coordination	-0.1369	-0.0540	0.1149

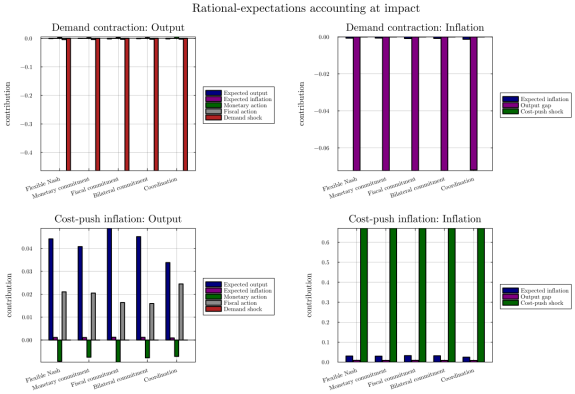
Cost-push inflation

Regime	m_0	f_0	Actual loss
Flexible Nash	0.2516	0.4091	0.5428
M persistence	0.2029	0.3989	0.5413
F persistence	0.2606	0.3182	0.5476
Bilateral	0.2098	0.3099	0.5460
Coordination	0.1934	0.4763	0.5325

Coordination is not weaker policy

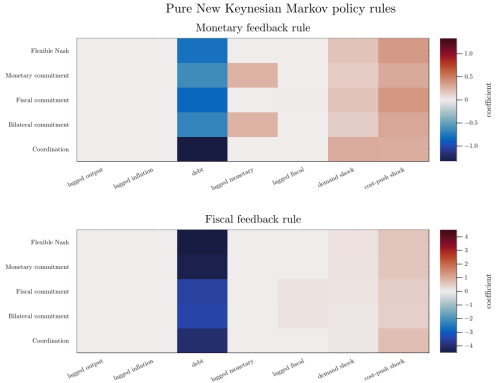
The planner reallocates stabilization across instruments under fixed social weights. Coordinated monetary policy can move more because the planner internalizes both agencies' spillovers.

How private expectations enter on impact



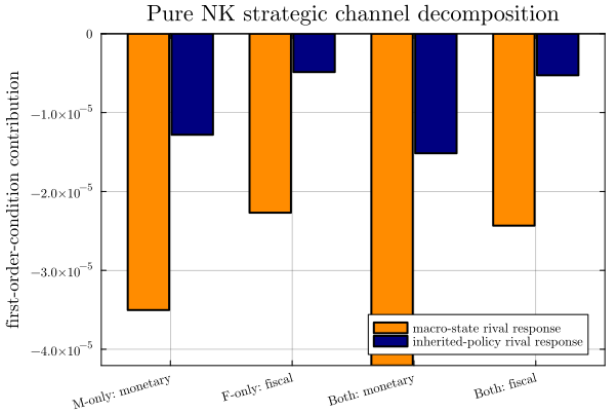
- The transitory demand innovation dominates current output.
- Following a cost-push shock, expected future output visibly contributes to current activity.

Policy persistence is not the same as leadership



- Own-lag coefficients are material when adjustment costs are active.
- Cross-lag coefficients are small: each authority binds its future self more strongly than it moves the rival.

The calibrated strategic-investment channel is small



Demand shock

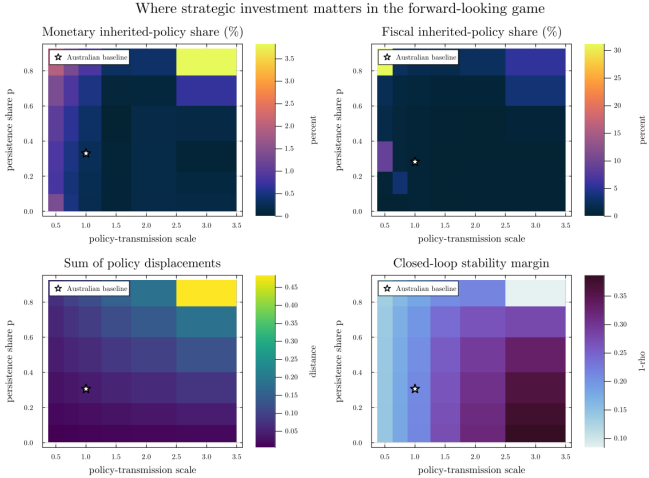
Inherited-policy bounded share:

- monetary persistence: 0.003;
- fiscal persistence: 0.002.

Diagnosis

Commitment capacity exists, but the rival's response to the inherited policy position is weak.

Where does the forward-looking strategic effect become large?



Read the parameter map jointly

Robust high-effect corner

At transmission scale $\tau = 3.00$ and $p = 0.85$:

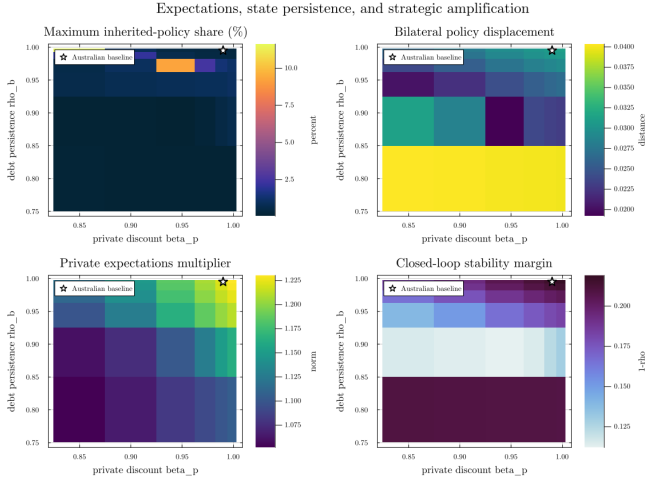
- policy displacement $= 0.484$;
- stability margin $= 0.084$;
- monetary inherited share $= 3.83\%$.

Strong transmission and durability reinforce each other, but stability thins.

High-share false positive

The fiscal share reaches 31.19% under weak transmission and high persistence, but policy displacement is only 0.071. Other continuation terms nearly cancel.

Expectations and debt persistence



Forward-lookingness enables strategy; it does not guarantee it

Monotonic object

The private expectations multiplier increases with β_p and reaches 1.230 in the grid.

Non-monotonic objects

The inherited-policy share and policy displacement do not move monotonically with β_p or ρ_b.

Economic reason

Anticipated stabilization can reduce the future shadow loss an authority wants to manipulate. Persistent debt can make both agencies react mainly to debt, rather than to one another's inherited instrument.

Expectations amplify an equilibrium reaction slope; they do not create that slope.

Equilibrium selection and fiscal dominance

Equilibrium search

- 7 dispersed initial forecasting maps;
- 1 distinct stable stationary solution;
- bilateral spectral radius 0.7834;
- all 72 parameter-map cells converge and are stable.

This is evidence, not a global uniqueness theorem.

Fiscal-dominance boundary

Debt creates fiscal pressure on monetary policy, but a full fiscal-dominance model requires nominal liabilities, a government valuation equation, and explicit monetary/fiscal backing regimes.

Conclusions

1. The static game identifies the key reaction slopes and distinguishes free riding from offset incentives.
2. Adjustment costs convert policy into a state, replacing imposed Stackelberg order with endogenous commitment capacity.
3. Feedback strategies are essential: open loop omits the rival-response component of strategic investment.
4. The NK block carries expected future policy back into current output and inflation through a rational-expectations fixed point.
5. In the Australian baseline, policy is persistent but the cross-policy reaction to inherited instruments is weak.

Empirical priority

Estimate the cross-policy response product—not only mandate weights or individual policy inertia.

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Questions and discussion

Backup: compact comparison of equilibrium concepts

Concept	What is fixed after a deviation?	Strategic object	Use here
Static Nash	rival's current action	within-period best response	benchmark conflict
Static Stackelberg	follower response function	imposed order of moves	benchmark leadership
Dynamic open loop	rival's announced future path	complete policy path	comparison only
Markov-perfect Nash	rival's state-contingent rule	endogenous state manipulation	main policy game
Coordination	one objective controls both instruments	social policy mix	welfare benchmark

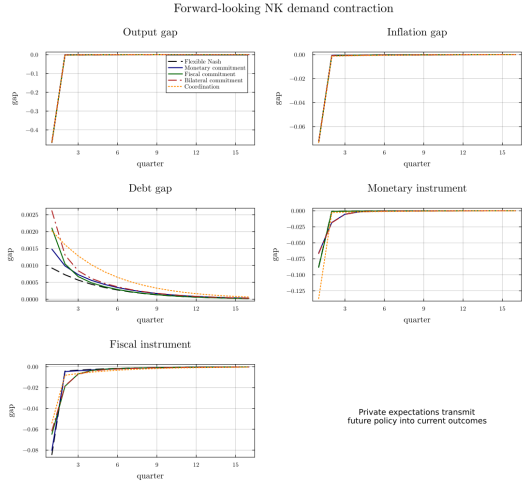
Backup: numerical algorithm

1. Initialize the private forecasting matrix G.
2. Construct $O(G)$ and reduced matrices A, B_M, B_F.
3. Solve the coupled policy Riccati equations for k_M, k_F, P_M, P_F.
4. Form the implied forecast map $\Phi(G)$.
5. Update $G \leftarrow (1-\delta)G + \delta\Phi(G)$ and repeat.
6. Verify private equations, FOCs, Bellman residuals, and $\rho(A_c) < 1$.

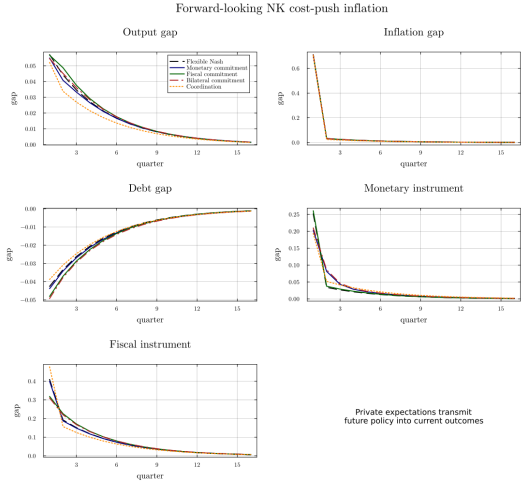
Baseline computation

The bilateral equilibrium takes 278 damped outer iterations. The Julia forward-looking suite contains 52 passing checks.

Backup: forward-looking demand-shock responses



Backup: forward-looking cost-push responses



Backup: welfare accounting

- **Macro loss**: common social inflation, output, debt, and instrument-level terms; excludes adjustment costs.
- **Actual loss**: adds the physical implementation cost of adjustment.

Demand experiment	Macro loss	Actual loss
Flexible Nash	0.1154	0.1154
Monetary persistence	0.1149	0.1153
Fiscal persistence	0.1151	0.1154
Bilateral persistence	0.1147	0.1153
Coordination	0.1149	0.1149

Interpretation

Inertia can improve macro stabilization while consuming real implementation resources. Coordination and commitment capacity remain distinct.

Backup: path toward a full fiscal-dominance model

- Replace the reduced-form debt law with a consolidated nominal government budget constraint.
- Introduce nominal maturity structure and the market value of debt.
- Specify monetary versus fiscal backing regimes.
- Let the price level participate in government liability valuation.
- Re-solve the game to determine whether strategic investment shifts the boundary between active monetary and active fiscal regimes.